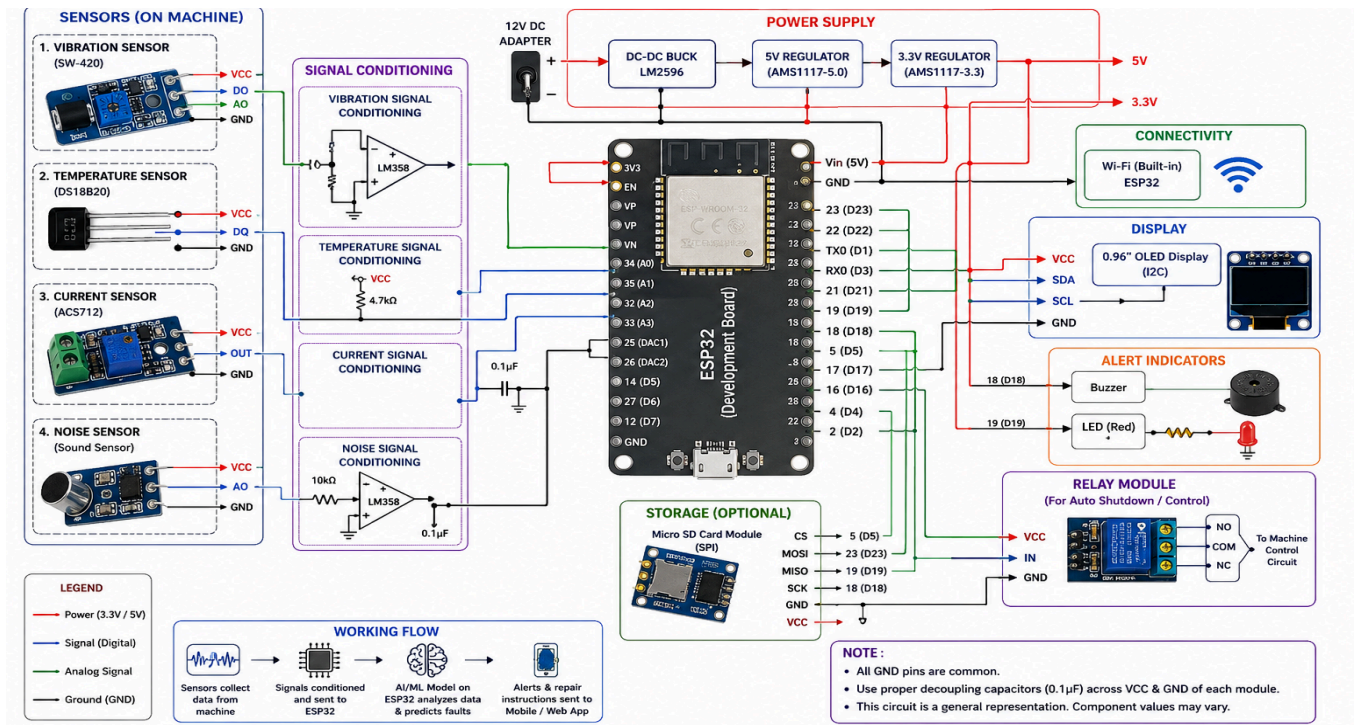


# AI-Based Machine Health Monitoring and Repair Assistant



The AI-Based Machine Health Monitoring and Repair Assistant is designed to continuously monitor the condition of industrial machines and predict possible faults before major breakdowns occur. The system uses vibration, temperature, current, and noise sensors installed on the machine to collect real-time operating data.

The sensor signals are sent through signal-conditioning circuits using components such as the LM358 and then provided to an ESP32 development board. The ESP32 processes the sensor data and uses its built-in Wi-Fi connectivity to send the information to an AI/ML model for analysis.

The AI system analyzes patterns in vibration, temperature, current, and noise to identify abnormal machine behavior and predict potential problems such as overheating, excessive vibration, bearing wear, or electrical abnormalities. Based on the analysis, the system generates a machine-health status and identifies whether maintenance may be required.

The results are displayed on a 0.96-inch OLED display and can also be sent to a mobile or web application through Wi-Fi. A buzzer and red LED provide alerts when abnormal conditions are detected. An optional microSD module can store sensor data for historical analysis, while a relay module can be used for automatic machine shutdown or control when required.

Overall, the system follows:

Sensors → Signal Conditioning → ESP32 → AI/ML Analysis → Fault Prediction → Alert → Repair Assistance.